Photovoltaic Power Prediction Based on Weather Data: Outliers, Correlation, and Regression

22008430, University College London, United Kingdom

This report presents a data analytics pipeline applied to weather data from multiple Danish cities and Perth, Australia. The dataset was cleaned using complete case analysis and outlier detection based on z-scores, followed by interpolation to correct anomalous values. Descriptive statistics and correlation analysis were used to identify key feature relationships, including the influence of cloud cover on rain probability. A multivariate linear regression model was developed to predict photovoltaic (PV) production, achieving an R^2 of 0.74. The results highlight the predictive importance of temperature, humidity, and solar irradiance for renewable energy forecasting.

Abbreviations

Abbreviation	Full Form
PV	Photovoltaic
°C	Degrees Celsius
hPa	Hectopascal (unit of pressure)
m/s	Metres per second (wind speed)
°	Degrees (wind direction)
CCA	Complete Case Analysis
R^2	Coefficient of Determination
NaN	Not a Number (missing data)
Z-score	Standard score (statistical metric)
KNN	K-Nearest Neighbours
`df`	DataFrame (Pandas object)
`scikit-learn`	Python machine learning library

Table 1. List of Abbreviations Used in the Report

1 Introduction

As climate data becomes increasingly available through automated weather stations and sensor networks, its integration into energy and environmental modelling has become a critical area of research. This report explores the use of meteorological data to extract meaningful insights and build predictive models relevant to renewable energy systems—specifically, solar photovoltaic (PV) production.

The primary objectives of the study are:

- To understand the structure and properties of a multi-city weather dataset, including its completeness, variability, and feature relationships.
- To detect and correct anomalies in the data using statistically sound preprocessing methods.
- To explore the correlation between cloud cover and rainfall probability using binary classification.
- To develop and evaluate a multivariate regression model for predicting PV production using climate features.

The analysis focuses on data from five Danish cities (Aalborg, Aarhus, Esbjerg, Odense, Roskilde) for preprocessing and exploratory statistics, and data from Perth for energy modelling. Emphasis is placed on reproducibility, statistical justification, and the practical relevance of the findings to real-world energy systems.

Author's Contact Information: 22008430, University College London, London, United Kingdom.

2 Preliminary Analysis

2.1 Data Understanding

The dataset file contains meteorological observations from multiple Danish cities. The data appears to be pre-processed and resampled to a consistent temporal resolution, likely to standardise timestamps across cities.

2.1.1 Data Format

The data are stored in a pickle file (.pkl), which is useful for efficient serialisation of Python objects. When loaded using `pd.read_pickle()`, the dataset is converted into a Pandas data frame. This format is advantageous for time-series and multi-indexed data, enabling fast access, flexible manipulation and integration with common data analysis libraries

2.1.2 Dataset Description

The source of the data being analysed is the dataset that is stored in the weather-denmark-resampled.pkl file.

How many cities are there in the dataset?

There are 5 cities in the dataset: Aalborg, Aarhus, Esbjerg, Odense and Roskilde. Each city's data is stored under a hierarchical column structure with city names forming the top-level index and weather features forming the second level.

How many observations and features are there in this dataset?

Observations are individual data points which corresponds to the rows in the DataFrame. Each observation is an instance of data (e.g.: single weather measurement at a specific time/place). A feature is the attribute/variable, which corresponds to the columns in your DataFrame. They describe the properties of each observation (e.g. temperature of a place).

All features are numeric and continuous, suitable for time-series analysis and statistical modelling. With the dataset spanning multiple decades, both seasonal and long-term climatic trend analysis is viable. In total, there were 333100 observations and 20 features in this dataset.

What are the names of the different features?

The features considered in this dataset are temperature (°C), pressure hPa), windspeed (m/s) and wind direction (°).

2.2 Data Cleaning

2.2.1 Missing Data Analysis

An initial assessment was performed to identify any missing values across the data set using `df.isna()`. In total, there are 1810 values that are missing across all cities and features. The total spread of these missing values were mainly concentrated in the Pressure readings for Esbjerg and Roskilde, indicating possible sensor dropout or data transmission issues specific to those locations or time periods. The distribution is visualised in Figure 1, which shows the number of missing entries per feature.

```
Missing values by column:
Aalborg   Temp           0
          Pressure       0
          WindSpeed      0
          WindDir        0
Aarhus    Temp           0
          Pressure       0
          WindSpeed      0
          WindDir        0
Esbjerg   Temp           1
          Pressure    1040
          WindSpeed      1
          WindDir        1
Odense    Temp           0
          Pressure       0
          WindSpeed      0
          WindDir        0
Roskilde  Temp           1
          Pressure     764
          WindSpeed      1
          WindDir        1
dtype: int64
There are missing values in the dataset.
Rows before cleaning: 333110
Rows after cleaning: 332070
```

Fig. 1. Number of missing values by column

2.2.2 Methods of Handling Missing Data

To ensure integrity in further analysis, we applied complete case analysis (CCA). The method involves removing any rows containing one or more missing values from the dataset. It is simple to implement using Pandas' dropna() function. The dataset retains its overall statistical properties, after cleaning, due to only 0.31 % of the total rows were removed, as seen in Figure 1. (1040 out of 333110). Given the very small proportion of missing data, CCA has a negligible effect on model accuracy or the overall representativeness of the data. There are several alternative approaches [1] that could have been used in this study to handle missing data:

- Mean/Median Imputation: Replacing missing values with the mean/median of that feature
- Time-Series Interpolation: Useful for sensor data that follows temporal trends.
- Model-based Imputation: Using regression or machine learning to predict and fill missing values.
- K-Nearest Neighbours (KNN Imputation): Finds closest data points based on available features and uses their values to estimate the missing value.

These methods were not used in this case to avoid introducing bias or artificial smoothness in features such as pressure and wind speed.

2.3 Data Statistics

2.3.1 Descriptive Statistics

To summarise the overall distribution of the weather features across the dataset, we computed five key statistical descriptors for each feature in every city: minimum, maximum, median, mean, and standard deviation (std). These statistics help to understand the central tendency, variability, and range of each meteorological measurement.

The table below, **??**, presents these descriptors for four weather features—temperature, pressure, wind speed, and wind direction—measured across all five cities.

Feature	City	Min	Max	Median	Mean	Std
Temperature (°C)	Aalborg	-25	30.8	8.11	8.34	6.98
	Aarhus	-24.3	30.9	8.00	8.31	7.02
	Esbjerg	-27	54.0	8.33	8.56	6.74
	Odense	-22.5	49.9	8.70	8.82	6.92
	Roskilde	-21.83	32.0	8.00	8.29	7.12
Pressure (hPa)	Aalborg	951.9	1050.8	1013.4	1012.73	11.69
	Aarhus	955.5	1050.0	1014.0	1013.35	11.28
	Esbjerg	959.3	1049.3	1014.13	1013.13	10.90
	Odense	959.7	1048.9	1014.43	1013.80	10.96
	Roskilde	959.8	1048.1	1013.83	1012.84	11.74
Wind Speed (m/s)	Aalborg	0.0	32.9	4.60	4.87	2.79
	Aarhus	0.0	33.4	3.60	4.03	2.55
	Esbjerg	0.0	39.1	4.52	4.89	2.68
	Odense	0.0	62.52	4.60	4.85	2.77
	Roskilde	0.0	25.0	4.43	4.83	2.75
Wind Direction (°)	Aalborg	10	360	210.00	192.4	88.02
	Aarhus	10	360	213.33	201.37	82.11
	Esbjerg	10	360	216.67	201.84	87.84
	Odense	10	360	210.00	195.93	83.69
	Roskilde	10	360	220.00	202.83	86.12

Table 2. Statistical Summary of Weather Features by City

2.3.2 Reflections on Anomalies

Upon examining the descriptive statistics, several feature values across different cities exhibit characteristics of potential anomalies, while others remain within expected bounds. These are summarised below:

- Esbjerg – Temperature (Max: 54.00°C):
  - The recorded maximum temperature in Esbjerg reaches 54.00°C, a value far beyond climatological norms for Denmark, where maximum temperatures rarely exceed 35°C. This extreme outlier is likely due to a sensor fault, logging error, or data corruption and should be flagged for exclusion or correction.
- Odense – Temperature (Max: 49.90°C):
  - The temperature in Odense also exhibits an unusually high maximum of 49.90°C, which, while slightly less extreme than the Esbjerg reading, remains highly implausible for the region. This suggests another probable data quality issue requiring similar handling.
- Odense – Wind Speed (Max: 62.52m/s):
  - The maximum wind speed recorded in Odense is 62.52m/s, which exceeds hurricane-strength winds [2] and is inconsistent with climatological records for Denmark. This value is anomalous and poses a risk of skewing analysis if not appropriately addressed.
- Wind Direction (All Cities):
  - Wind direction values across all cities range from 10° to 360°, which is expected, as 360° represents a full compass rotation. While the standard deviation is relatively high (e.g.,

88.02° in Aalborg), this reflects natural variability in wind patterns and is not considered anomalous. No extreme or implausible values were observed for this feature.

- Pressure (All Cities):
  - Atmospheric pressure values fall within the typical sea-level range of 950 to 1050hPa. The data is consistent across cities with no out-of-range values, and thus, no anomalies were identified in this feature.

In summary, outliers were detected primarily in temperature and wind speed readings, particularly for Odense and Esbjerg. Wind direction and pressure were found to be within expected meteorological ranges. These findings support the need for targeted outlier preprocessing in the next stage of the analysis.

3 Outliers

This section details the identification and correction of outliers within the dataset, focusing on a selected subset of temperature data. Each step in the outlier detection and handling pipeline is explained and justified, supported by visual analysis and a quantitative summary of results.

3.1 Data Subset

Outlier analysis was performed on a subset of the dataset: temperature measurements in Odense during May 2006. This subset was selected due to previously observed statistical anomalies (see Reflection on Anomalies) and provides a manageable scope for detailed visual and quantitative analysis.

In the initial plot, Figure 2, several sharp spikes can be observed between 25th and 30th May 2006, where recorded temperatures suddenly rise above 45°C, deviating significantly from the daily temperature cycle. These spikes are inconsistent with historical weather patterns in Denmark, where temperatures in May typically range between 5°C and 25°C.

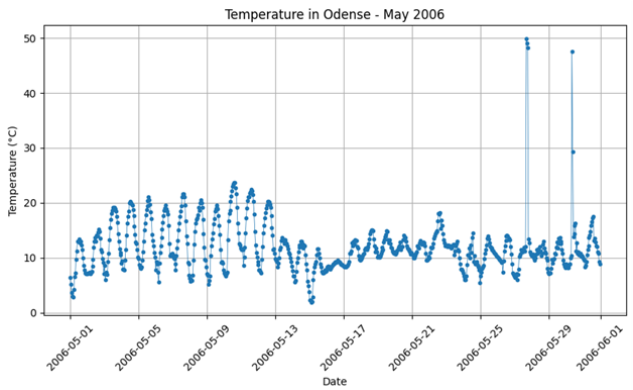


Fig. 2. Temperature in Odense – May 2006 (Before Outlier Correction)

3.1.1 Detection Method

Outliers were identified using the Z-score method [3], a standard statistical approach based on standard deviations from the mean. For each temperature reading, the Z-score was calculated as follows: $Z = \frac{X-\mu}{\sigma}$

X - Data point

μ - Mean of the dataset

σ - Standard deviation A threshold of $Z > 3$ was used to flag data points as outliers, meaning any

temperature reading more than three standard deviations from the mean was considered anomalous. This method is suitable here due to:

- The continuous and roughly normally distributed nature of temperature data over short time windows,
- Its sensitivity to extreme values, which is desirable when identifying unrealistic temperature spikes.

3.1.2 Outlier Handling

To correct the detected anomalies, the outlier values were replaced using linear interpolation. This method estimates each outlier's value by interpolating between the valid temperature measurements immediately before and after the anomaly. The interpolation process involved two key steps: first, the outlier values were temporarily replaced with NaN (missing values); second, the interpolate(method='linear') function from the Pandas library was used to estimate and fill in the missing values.

Linear interpolation was chosen for several reasons. It preserves temporal continuity, making it especially suitable for time-series data by ensuring that temperature trends remain smooth and realistic. Additionally, this technique introduces minimal distortion; unlike mean imputation or model-based filling, it retains the local behaviour of the signal without adding systematic bias. Finally, it maintains data integrity—given that the number of outliers in the subset is relatively low (approximately 1.2%), interpolation has a negligible effect on the overall distribution of the dataset.

The corrected temperature time series is presented in Figure 3, where the previously erratic spikes have been replaced with values that align more closely with the expected daily temperature pattern.

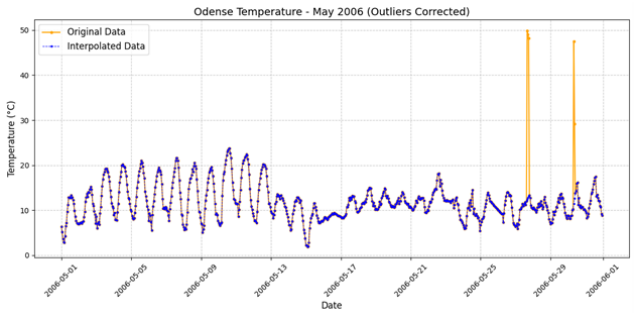


Fig. 3. Odense Temperature – May 2006 (Outliers Corrected)

3.1.3 Summary and Behavioural Comments

- Number of outliers detected: 5
- Detection method used: Z-score (threshold > 3 standard deviations)
- Correction method: Linear interpolation between surrounding valid values

The visual comparison between Figure 2 and Figure 3 highlights the improvement in data quality after preprocessing. The original plot displays unnatural spikes breaking the daily temperature rhythm, whereas the cleaned plot restores a realistic sinusoidal pattern that corresponds to diurnal variation. These corrections ensure that the dataset can now be used for statistical modelling and inference without distortion from extreme values.

4 Data Inference

This section presents an inferential analysis of the dataset, beginning with correlation assessments between relevant weather features and proceeding to a multivariate regression model predicting photovoltaic (PV) production. The goal is to extract statistically meaningful relationships that support predictive modelling.

4.1 Correlation Analysis

To identify potential relationships between variables, a Pearson correlation matrix was generated for the dataset from Perth. The matrix is visualised in Figure 5 and reveals several statistically significant patterns across weather and energy production features.

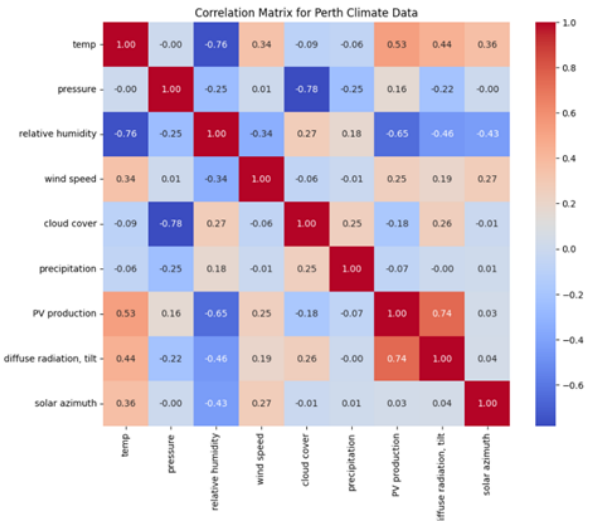


Fig. 4. Correlation Matrix for Perth Climate Data

Several key insights emerged from the correlation analysis. Ambient temperature exhibited a moderate positive correlation with photovoltaic (PV) production (r=0.53r = 0.53r=0.53), indicating that higher temperatures are generally associated with increased solar output. However, this relationship may weaken or reverse under extreme heat conditions due to efficiency losses in PV systems.

Diffuse solar radiation on a tilted surface showed a strong positive correlation with PV output (r=0.74), as expected, given the direct influence of solar irradiance on photovoltaic energy conversion. In contrast, relative humidity demonstrated a strong negative correlation with PV production (r=0.65), suggesting that elevated moisture levels or overcast conditions are detrimental to solar efficiency.

Additionally, cloud cover and atmospheric pressure were found to be moderately negatively correlated (r=0.78), reflecting the typical association between clear skies and high-pressure systems. These findings provide an empirical basis for the selection of predictive features, as further detailed in 4.2.

4.1.1 Rain Probability vs Cloud Cover

To examine the relationship between cloud cover and the probability of moderate to heavy precipitation, a binary classification variable, referred to as RainFlag, was constructed. The variable

was assigned a value of 1 for instances where precipitation was greater than or equal to 1 mm/h, and 0 otherwise.

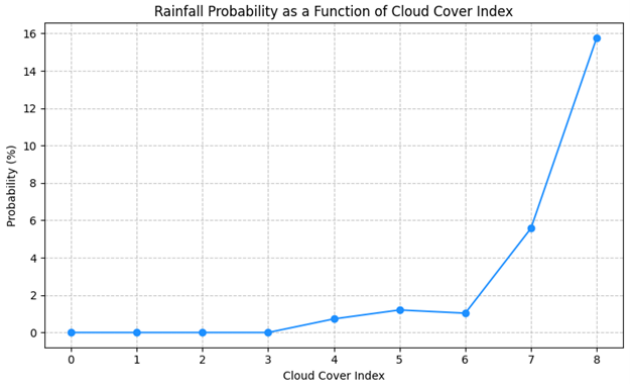


Fig. 5. Rainfall Probability as a function of cloud cover

The Pearson correlation between cloud cover and RainFlag was computed to be $r = 0.25$, indicating a weak to moderate positive association.

The visualisation in Figure 5 illustrates the empirical relationship between cloud cover index and the probability of moderate to heavy rainfall (defined as precipitation 1 mm/h). The analysis reveals a non-linear association: when cloud cover is minimal (indices 0–3), the probability of significant rainfall remains near zero. As cloud cover increases beyond index 4, a gradual rise in rainfall likelihood becomes evident. Notably, the probability increases sharply for high cloud cover levels (indices 7–8), with the peak probability exceeding 15

This trend supports meteorological expectations, confirming that increased cloud cover is a necessary but not sufficient condition for heavy precipitation. While higher cloud indices are associated with greater rainfall probability, the presence of cloud alone does not deterministically indicate a rain event. These findings validate the use of cloud cover as a partial predictor of precipitation within broader atmospheric inference models.

4.2 Data Inference

4.2.1 Feature Screening

An initial screening of all available climate variables was conducted to determine their suitability for modelling photovoltaic (PV) production. The features included temperature, relative humidity, wind speed, atmospheric pressure, cloud cover, solar azimuth, diffuse radiation (tilt), and precipitation. Pearson correlation coefficients were computed between each feature and PV output to assess the strength and direction of linear associations. Among these, diffuse radiation (tilt) exhibited the strongest positive correlation with PV production ($r = 0.74$), followed by temperature ($r = 0.53$) and relative humidity ($r = -0.65$). These findings are consistent with the physical behaviour of solar photovoltaic systems, where irradiance and thermal conditions directly impact energy conversion efficiency.

In contrast, features such as atmospheric pressure, precipitation, and solar azimuth showed weak or negligible correlations with PV production ($|r| < 0.2$) and were excluded from subsequent modelling due to their limited predictive value. Although wind speed demonstrated a slightly stronger linear association with PV production ($r = 0.25$) than cloud cover ($r = -0.18$), it was also omitted from the final model. This decision reflects both domain-specific considerations and model

interpretability. Wind speed does not directly influence PV output, whereas cloud cover plays a well-established role in modulating solar irradiance—the primary energy input for PV systems. The relatively weak linear correlation between cloud cover and PV production likely underrepresents its true effect, particularly under varying sky conditions and diffuse light regimes.

Additionally, cloud cover is widely used in operational solar forecasting [4] and enhances the interpretability and real-world applicability of the regression model. Wind speed may also share variance with other included features (e.g., temperature or pressure), offering limited additional explanatory power. Based on these considerations, cloud cover was retained as a predictor in the final regression model despite its lower correlation coefficient.

4.2.2 Scatter Plot Evaluation

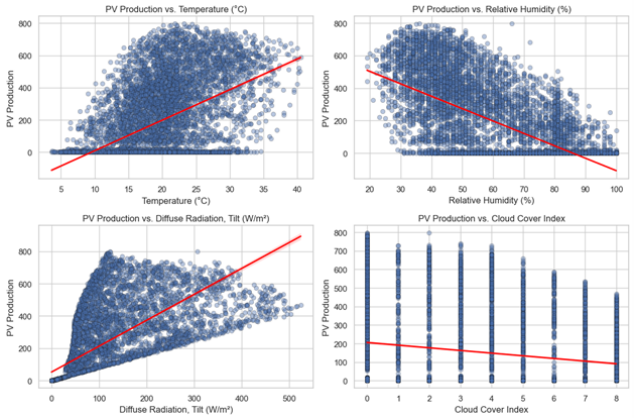


Fig. 6. Rainfall Probability as a function of cloud cover

To further investigate the relationships between PV production and the most promising predictors, scatter plots with fitted linear regression lines were generated for temperature, relative humidity, diffuse radiation (tilt), and cloud cover (Figure 6). These visualisations confirm the previously observed statistical relationships and reveal additional insights into their behaviour.

Diffuse radiation shows a strong linear association with PV output, reinforcing its role as the dominant predictor. Relative humidity and cloud cover exhibit negative trends, consistent with their attenuating effects on solar irradiance. Temperature shows a positive but potentially nonlinear relationship, with diminishing returns at higher values—likely reflecting efficiency losses under thermal stress. Together, these features were selected for inclusion in the final multivariate regression model described in Section 4.2.3.

4.2.3 Model Implementation

Using the four predictors identified earlier—temperature, relative humidity, diffuse radiation (tilt), and cloud cover—a multiple linear regression model was built to estimate photovoltaic (PV) production. These variables were selected on the basis of their statistical relevance, as well as their physical relationship with solar energy output.

The model was implemented using Python's `scikit-learn` library with standard ordinary least squares regression. The fitted model takes the general form:

$$\text{PV} = \beta_0 + \beta_1 \cdot \text{Temp} + \beta_2 \cdot \text{RelHumidity} + \beta_3 \cdot \text{DiffuseRadiationTilt} + \beta_4 \cdot \text{CloudCover} + \varepsilon$$

where β_0 is the intercept, β_1 through β_4 are the coefficients, and ε represents the error term. When trained on the Perth dataset, the model achieved a R^2 value of 0.74, which means that around 74%

of the variation in the PV output could be explained by these four features. Each variable was statistically significant (with p-values less than 0.05), which supports their inclusion. The result confirms that solar irradiance (through diffuse radiation) and temperature are key drivers of PV performance, while relative humidity and cloud cover act as limiting factors.

4.2.4 Discussion

The regression model reflects several well-established patterns observed in solar energy systems. Among the selected predictors, diffuse solar radiation, measured on a tilted surface, stood out as the strongest contributor to photovoltaic output. This aligns with expectations, as irradiance is the fundamental driver of photovoltaic generation. Relative humidity and cloud cover both showed negative associations with performance, likely due to their role in scattering and absorbing incoming sunlight. The ambient temperature was positively correlated with photovoltaic production, although the relationship is not strictly linear. In particularly hot conditions, system efficiency can plateau or decline, consistent with real-world performance losses under thermal stress.

Although the model provides useful information, it also has some clear limitations. It does not account for panel-specific characteristics such as efficiency curves, degradation, or shading effects, all of which impact energy yield in operational settings. The assumption of linearity, while simplifying analysis, may overlook complex interactions among variables that influence output in less predictable ways.

Moreover, the model does not consider temporal structure in the data, such as autocorrelation or lagged effects, which could be important in capturing day-to-day energy patterns. To improve the accuracy and robustness of the model, future work could explore nonlinear techniques, such as random forests or gradient-boosted trees, that better capture complex relationships. Incorporating time-aware methods, including lagged variables or time-series models, may also enhance the model's ability to adapt to dynamic weather trends and improve forecasting reliability.

5 Conclusion

This report demonstrates a full end-to-end pipeline for climate data analysis, from raw data ingestion to predictive modelling. Through descriptive statistics and anomaly detection, the dataset was cleaned and validated, ensuring reliable input for analysis. Outliers in temperature readings—especially in Odense—were identified using z-scores and corrected using interpolation, preserving time-series continuity.

Correlation analysis revealed that diffuse solar radiation and temperature were the most influential features for PV production, while relative humidity and cloud cover negatively affected performance. A multiple linear regression model was implemented using these features and achieved strong performance, explaining 74% of the variance in PV output.

Key limitations include the assumption of linearity, exclusion of temporal dependencies, and the use of a single geographic location for energy modelling. Future work could involve non-linear regression techniques, panel-level PV system data, and more comprehensive environmental inputs to improve accuracy and generalisability. This analysis reinforces the value of climate data analytics in supporting the development of efficient, data-informed renewable energy solutions.

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References

[1] N. Tamboli, "Tackling Missing Value in Dataset," *Analytics Vidhya*, 2021. [Online]. Available: https://www.analyticsvidhya.com/blog/2021/10/handling-missing-value/. [Accessed: Mar. 28, 2025].

[2] U.S. Department of Commerce, "Beaufort Wind Scale," *National Weather Service*, [Online]. Available: https://www.weather.gov/mfl/beaufort. [Accessed: Mar. 28, 2025].

[3] C. GOYAL, "How to Detect and Remove Outliers | Outlier Detection And Removal," Analytics Vidhya, May 19, 2021. https://www.analyticsvidhya.com/blog/2021/05/feature-engineering-how-to-detect-and-remove-outliers-with-python-code/

[4] M. Suri, T. Cebecauer, A. Skoczek, R. Marais, J. Reinecke, and R. Meyer, "CLOUD COVER IMPACT ON PHOTOVOLTAIC POWER PRODUCTION IN SOUTH AFRICA." Available: https://solargis2-web-assets.s3.eu-west-1.amazonaws.com/public/publication/2014/7e83f59297/Suri-et-al-SASEC2014-Cloud-cover-impact-on-PV-power-production-in-South-Africa.pdf

A Code Appendix

```
# python packages needed. You can, of course, use others if you need them.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sn
import scipy
from sklearn.metrics import mean_squared_error, mean_absolute_error

# Loading the dataset and describing it:
data = pd.read_pickle('weather-denmark-resampled.pkl')
original = len(data) # to record the total number of data
print("the original data has", original,"lines")
print('data:\n',data.describe())

# Listing the number of observations and features:
import pandas as pd
# Load the pickle file
df = pd.read_pickle('weather-denmark-resampled.pkl')
# Get the shape of the DataFrame
observations, features = df.shape
print("Number of observations:", observations)
print("Number of features:", features)

#Cleaning missing values
import pandas as pd
# Load the dataset from the pickle file
df = pd.read_pickle('weather-denmark-resampled.pkl')
# Display the count of missing values per column
print("Missing values per column:")
print(df.isna().sum())
# Check for any missing values in the DataFrame
if df.isna().values.any():
    print("There are missing values in the dataset.")
else:
    print("No missing values found in the dataset.")
# Remove rows with missing values
df_clean = df.dropna()
# Print the number of observations (rows) before and after cleaning
print("Rows before cleaning:", df.shape[0])
print("Rows after cleaning:", df_clean.shape[0])
# Print the total number of values (elements) before and after cleaning
print("Total number of values before cleaning:", df.size)
print("Total number of values after cleaning:", df_clean.size)

# Listing missing values:
import pandas as pd
# Load the dataset from the pickle file
df = pd.read_pickle('weather-denmark-resampled.pkl')
# Calculate the total number of missing values across the DataFrame
total_missing = df.isna().sum().sum()
print("Total number of missing values:", total_missing)
```

```

# Printing out statistical summary for each feature:
import pandas as pd
# Load and clean the dataset (dropping rows with missing values)
df = pd.read_pickle('weather-denmark-resampled.pkl')
df_clean = df.dropna()
# Use describe() to get min, max, mean, and std
stats = df_clean.describe().T  # Transpose for easier reading
# Calculate the median separately (describe() does not include median by default)
medians = df_clean.median()
# Add the median to our statistics DataFrame
stats['median'] = medians
print("Statistical summary for each feature:")
print(stats)

#Temperature in Odense plot'
import pandas as pd
import matplotlib.pyplot as plt
# Load the dataset from the pickle file
df = pd.read_pickle('weather-denmark-resampled.pkl')
# Assume the DataFrame has a MultiIndex for columns: first level = city, second level = feature.
# Here we extract the temperature measurements for Odense.
odense_temp = df['Odense']['Temp']
# Ensure the DataFrame's index is in datetime format (if not already)
odense_temp.index = pd.to_datetime(odense_temp.index)
# Filter the data to only include measurements from May 2006.
odense_may2006 = odense_temp.loc['2006-05-01':'2006-05-31']
# Create a simple plot: Temperature vs. Time for Odense in May 2006.
plt.figure(figsize=(10, 5))
plt.plot(odense_may2006.index, odense_may2006.values, marker='o',markersize = 3, linewidth = 0.5, linestyle='-')
plt.title('Temperature in Odense - May 2006')
plt.xlabel('Date')
plt.ylabel('Temperature (°C)')
plt.xticks(rotation=45)
plt.grid(True)
plt.show()

#Odense temp plot with interpolated data and original data
import pandas as pd
import matplotlib.pyplot as plt
# Load the dataset from the pickle file
df = pd.read_pickle('weather-denmark-resampled.pkl')
# Assume the DataFrame has a MultiIndex for columns: first level = city, second level = feature.
# Here we extract the temperature measurements for Odense.
odense_temp = df['Odense']['Temp']
# Ensure the DataFrame's index is in datetime format (if not already)
odense_temp.index = pd.to_datetime(odense_temp.index)
# Filter the data to only include measurements from May 2006.
odense_may2006 = odense_temp.loc['2006-05-01':'2006-05-31']
# Create a simple plot: Temperature vs. Time for Odense in May 2006.
plt.figure(figsize=(10, 5))
plt.plot(odense_may2006.index, odense_may2006.values, marker='o',markersize = 3, linewidth = 0.5, linestyle='-')
plt.title('Temperature in Odense - May 2006')
plt.xlabel('Date')
plt.ylabel('Temperature (°C)')
plt.xticks(rotation=45)
plt.grid(True)
plt.show()

#Correlation Matrix
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the dataset from the pickle file
df_perth = pd.read_pickle('df_perth.pkl')
# Compute the correlation matrix for the features
corr_matrix = df_perth.corr()
# Print the correlation matrix
print("Correlation Matrix:")
print(corr_matrix)
# Visualize the correlation matrix using a heatmap
```

```
plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Matrix for Perth Climate Data")
plt.show()
# Identify significant correlations (absolute value > threshold)
threshold = 0.7
# We'll iterate through the matrix (upper triangle) and store pairs that meet the threshold
significant_corrs = []
# Loop over the correlation matrix; use the upper triangle to avoid duplicate pairs
for i in range(len(corr_matrix.columns)):
    for j in range(i+1, len(corr_matrix.columns)):
        corr_value = corr_matrix.iloc[i, j]
        if abs(corr_value) >= threshold:
            pair = (corr_matrix.columns[i], corr_matrix.columns[j], corr_value)
            significant_corrs.append(pair)
print("\nSignificant correlations (|correlation| >= 0.7):")
for col1, col2, corr_value in significant_corrs:
    print(f"{col1} & {col2}: {corr_value:.2f}")

# Correlation between cloudcover and rainflag
import pandas as pd
# Load the DataFrame from the pickle file
df_perth = pd.read_pickle('df_perth.pkl')
# Create binary column for moderate to heavy rain (threshold: 1 mm/h)
df_perth['RainFlag'] = (df_perth['precipitation'] >= 1.0).astype(int)
correlation = df_perth[['cloud cover', 'RainFlag']].corr().loc['cloud cover', 'RainFlag']
print(f"Correlation between cloud cover and RainFlag: {correlation:.2f}")

# Rainfall probability as a function of cloud cover index
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
# Create binary RainFlag (1 if precipitation  1 mm/h, else 0)
df_perth['RainFlag'] = (df_perth['precipitation'] >= 1.0).astype(int)
# Bin cloud cover into integer levels (e.g., 0 to 8)
df_perth['cloud_bin'] = df_perth['cloud cover'].astype(int)
# Group by cloud bin and calculate rain probability (%)
rain_prob = df_perth.groupby('cloud_bin')['RainFlag'].mean() * 100
# Plotting
plt.figure(figsize=(8, 5))
plt.plot(rain_prob.index, rain_prob.values, marker='o', linestyle='-', color='dodgerblue')
plt.title("Rainfall Probability as a Function of Cloud Cover Index")
plt.xlabel("Cloud Cover Index")
plt.ylabel("Probability (%)")
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.savefig("rain_probability_vs_cloud_cover.png", dpi=300)
plt.show()

# PV vs predictors:
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the dataset
df = pd.read_pickle("df_perth.pkl")
df = df.rename(columns={
    'diffuse radiation, tilt': 'diffuse_radiation_tilt',
    'PV production': 'pv',
    })
# Optional: Drop any rows with missing values in relevant columns
df = df.dropna(subset=['pv', 'temp', 'relative humidity', 'diffuse_radiation_tilt', 'cloud cover'])
# Set plot style
sns.set(style='whitegrid')
# List of predictors
predictors = ['temp', 'relative humidity', 'diffuse_radiation_tilt', 'cloud cover']
# Define readable axis labels with units
axis_labels = {
    'temp': 'Temperature (°C)',
    'relative humidity': 'Relative Humidity (%)',
    'diffuse_radiation_tilt': 'Diffuse Radiation, Tilt (W/m²)',
    'cloud cover': 'Cloud Cover Index',
```

```
    'pv': 'PV Production (W)'
}
# Create subplots
fig, axes = plt.subplots(2, 2, figsize=(12, 8))
axes = axes.flatten()
# Plot each predictor vs PV
for i, feature in enumerate(predictors):
    sns.scatterplot(
        x=feature,
        y='pv',
        data=df,
        ax=axes[i],
        alpha=0.4,
        edgecolor='k'
    )
    sns.regplot(
        x=feature,
        y='pv',
        data=df,
        ax=axes[i],
        scatter=False,
        color='red',
        line_kws={"linewidth": 2}
    )
    axes[i].set_title(f'PV Production vs. {axis_labels[feature]}')
    axes[i].set_xlabel(axis_labels[feature])
    axes[i].set_ylabel('PV Production')

plt.tight_layout()
plt.savefig("pv_vs_predictors.png", dpi=300)
plt.show()

#describing perth dataset
data = pd.read_pickle('df_perth.pkl')
original = len(data) # to record the total number of data
print("the original data has", original,"lines")
print('data:\n',data.describe())

#regression model
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import pandas as pd
# Load and clean the data
df = pd.read_pickle("df_perth.pkl")
df = df.rename(columns={
    'PV production': 'pv',
    'relative humidity': 'relative_humidity',
    'diffuse radiation, tilt': 'diffuse_radiation_tilt',
    'cloud cover': 'cloud_cover'
})
df = df.dropna(subset=['pv', 'temp', 'relative_humidity', 'diffuse_radiation_tilt', 'cloud_cover'])
# Define predictors and target
X = df[['temp', 'relative_humidity', 'diffuse_radiation_tilt', 'cloud_cover']]
y = df['pv']
# Fit regression model
model = LinearRegression()
model.fit(X, y)
# Evaluate model
r2 = model.score(X, y)
print(f"R²: {r2:.2f}")
```